Translations of functions

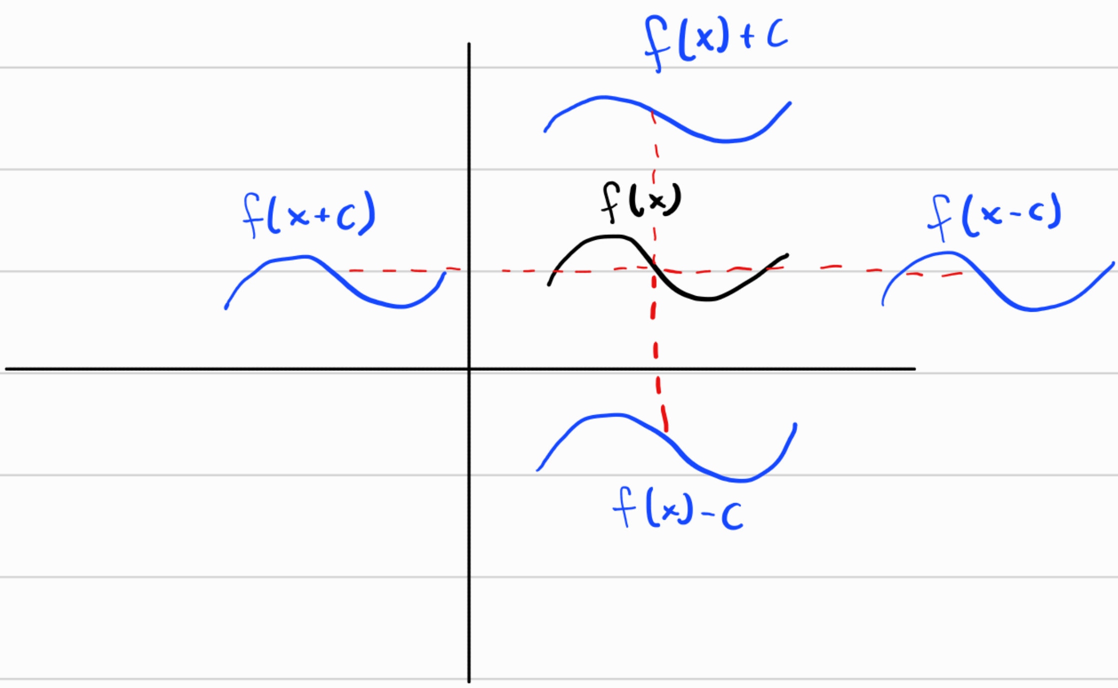
There are a few ways to make new functions, we start with the most straight-forward called translations.

Translations are just when we pull a function to the left, right, up, or down. This shift is done in a specific way:

let $c > 0$, $f(x)$ a function.

- $y = f(x) + c$, Shifts $f(x)$ up c units
- $y = f(x) - c$ Shifts $f(x)$ down c units
- $y = f(x+c)$ Shifts $f(x)$ left c units
- $y = f(x-c)$ Shifts $f(x)$ right c units

Notice the last two are a bit unintuitive.



Streching and reflecting

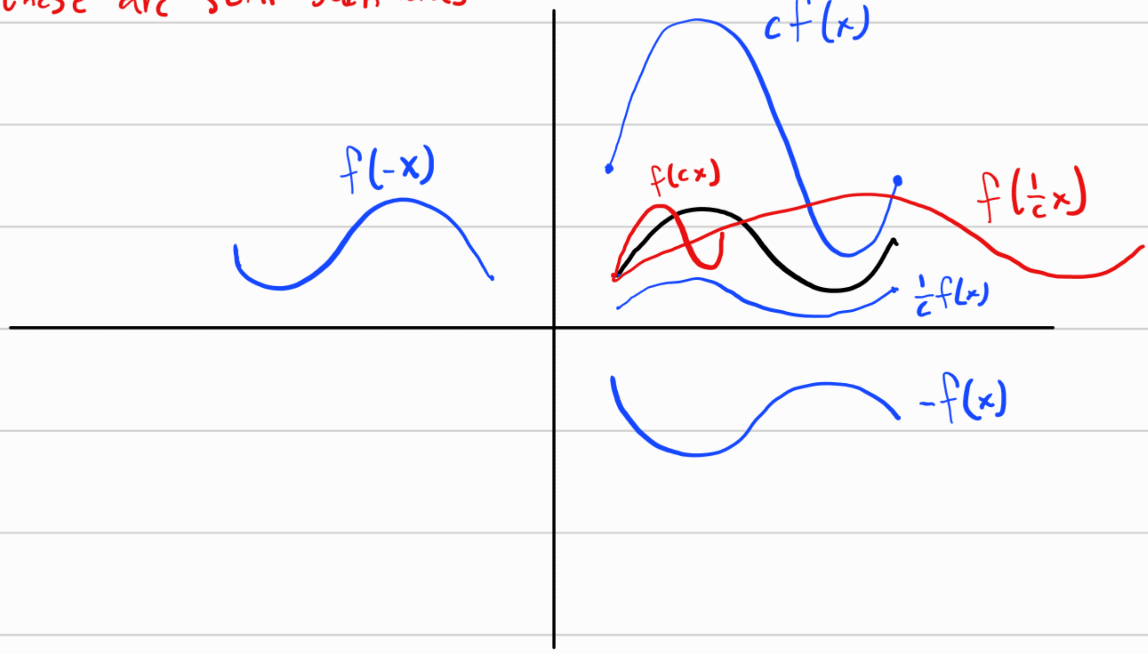
Another way to modify functions is by streching, shrinking or reflection over x or y axis.

This is accomplished as follows:

let $c > 1$ and $f(x)$ a function.

- $y = cf(x)$, Strech $f(x)$ Vertically by c.
- $y = \frac{1}{c}f(x)$, Shrink $f(x)$ Vertically by c.
- $y = f(cx)$, Shrink $f(x)$ horizontally by c.
- $y = f(\frac{1}{c}x)$, Strech $f(x)$ horizontally by c.
- $y = -f(x)$, reflect $f(x)$ over x-axis.
- $y = f(-x)$, reflect $f(x)$ over y-axis.

Notice these are still backwards



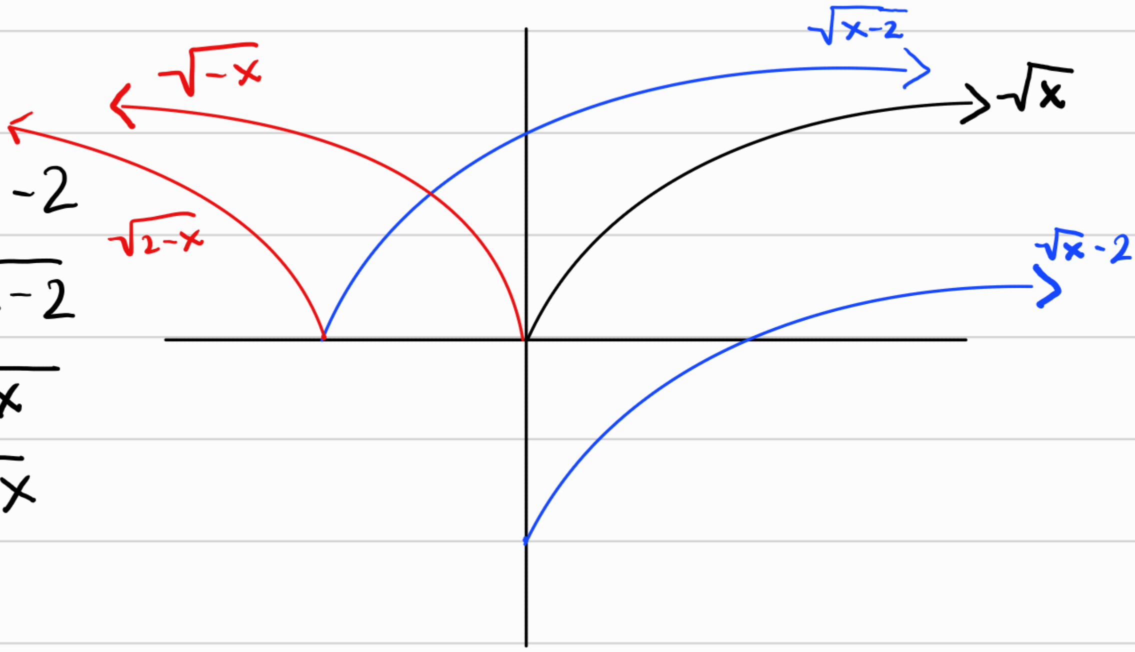
Ex:

$y=\sqrt{x}\ -2$

$y=\sqrt{x-2}$

$y=\sqrt{-x}$

$y=\sqrt{2-x}$



Ex:

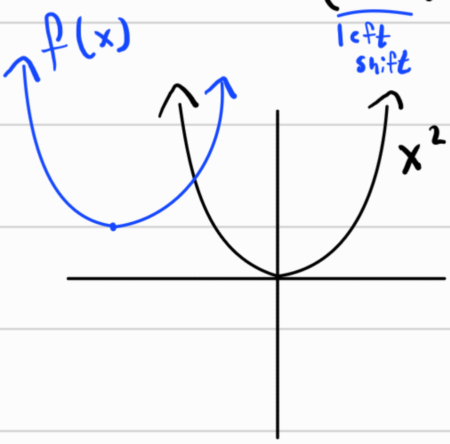
$$9=\left(\frac{6}{2}\right)^2$$

$$f(x)=x^2+6x+10$$

$$=x^2+6x+9-9+10$$

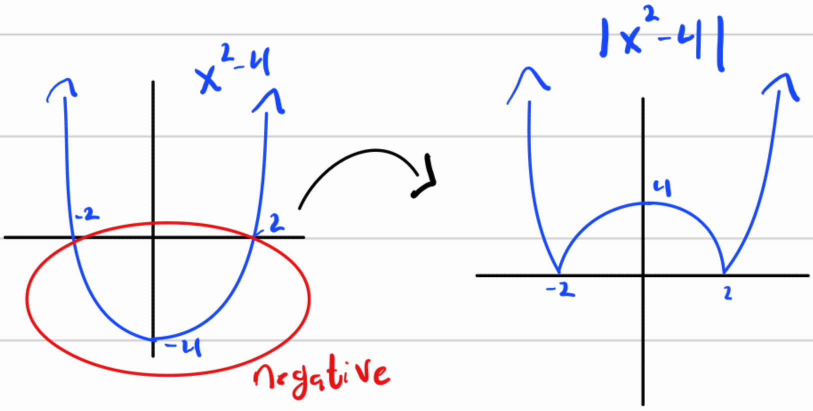
$$=\underbrace{(x+3)^2}_{\text{left shift}}\ \underbrace{+1}_{\text{Vertical shift}}$$

$-x^2$



Another common opperation is absolute value, it just makes negatives positive.

Ex: $f(x)=|x^2-4|$



Combining functions - algebraic methods

The usual four algebraic opperations (add, sub, mult, div) still works between two functions and yeild new functions.

Given two functions $f(x)$ and $g(x)$, the following are also functions:

- Sum: $(f+g)(x) = f(x) + g(x)$
- difference: $(f-g)(x) = f(x) - g(x)$
- Product: $(fg)(x) = f(x)g(x)$
- Quotient: $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$

the domain of each of the above is

$\mathrm{dom}(f) \cap \mathrm{dom}(g)$ (Possibly also removing division by 0)

that is, the domain of the new function is where both $f(x)$ and $g(x)$ are defined and we don't divide by 0.

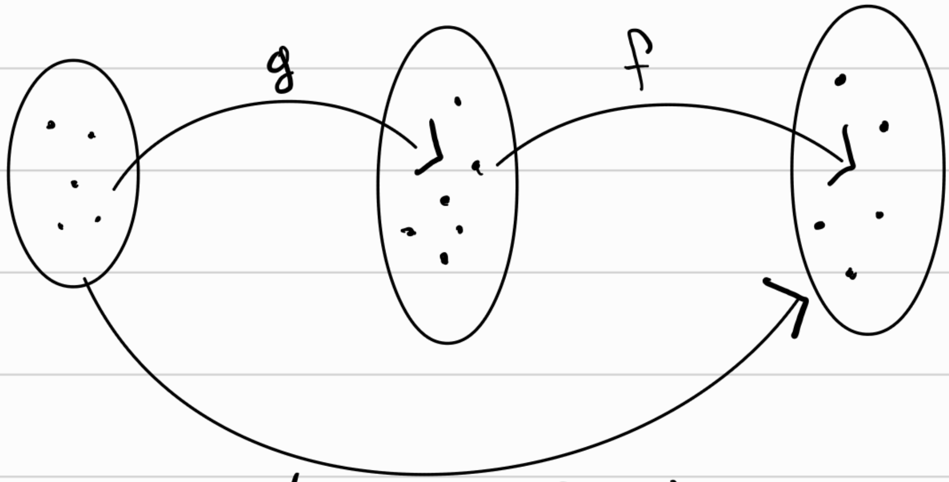
Ex: $f(x) = \sqrt{x}$, $g(x) = x-3$

$\left(\frac{f}{g}\right)(x) = \frac{\sqrt{x}}{x-3}$, $\left(\frac{g}{f}\right)(x) = \frac{x-3}{\sqrt{x}}$

$D = [0,3) \cup (3,\infty)$ $D = (0,\infty)$

Function Composition

All of our functions take real numbers and output real numbers. What if we take the output of one function as the input of another?



$$(f\circ g)(x) = f(g(x))$$

This is called function composition. Notice how $g(x)$ happens first.

Given two functions f and g, the composite function, (often called composition) is defined by

$$(f\circ g)(x) = f(g(x))$$

"f of g of x"

Domain is hard to think about here, technically domain$(f\circ g)$ is $\{x\in \text{dom}(g) \mid g(x)\in \text{dom}(f)\}$.

Usually, we find domain last in such functions.

Ex: $f(x) = x^2$, $g(x) = x-3$

$$(f \circ g)(x) = f(g(x)) = [g(x)]^2 = (x-3)^2 = x^2 - 6x + 9$$

$$(g \circ f)(x) = g(f(x)) = f(x) - 3 = x^2 - 3$$

In general, $f \circ g \neq g \circ f$

Ex: $f(x) = \sqrt{x}$, $g(x) = \sqrt{2-x}$

- $(f \circ f)(x) = f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = \sqrt[4]{x}$
  $D = [0, \infty)$

- $(f \circ g)(x) = \sqrt{g(x)} = \sqrt{\sqrt{2-x}} = \sqrt[4]{2-x}$
  $D = (-\infty, 2]$

- $(g \circ f)(x) = g(f(x)) = \sqrt{2 - f(x)} = \sqrt{2 - \sqrt{x}}$
  $D = [0, 4]$

- $(g \circ g)(x) = g(g(x)) = \sqrt{2 - g(x)} = \sqrt{2 - \sqrt{2-x}}$
  $D = [-2, 2]$

We can also compose more than 2 functions

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

Ex: $f(x) = \dfrac{x}{x+1}$, $g(x) = x^{10}$, $h(x) = x+3$

$$(f \circ g \circ h)(x) = f(g(h(x))) = \frac{g(h(x))}{g(h(x))+1} = \frac{[h(x)]^{10}}{[h(x)]^{10}+1}$$

$$= \frac{(x+3)^{10}}{(x+3)^{10}+1} \qquad D = \mathbb{R}$$

We can also <u>decompose</u> functions

Ex: $F(x) = \cos^2(x+9) = (f \circ g \circ h)(x)$, find $f(x), g(x), h(x)$

$\hookrightarrow \cos^2(x) = [\cos(x)]^2$

$$\boxed{\begin{array}{l} f(x) = x^2, \\ g(x) = \cos(x) \\ h(x) = x+9 \end{array}} \quad (g \circ h)(x) = \cos(x+9)$$